

Homework – Graduate Course in Mechatronic systems

1) Choose a **nonlinear** process that can be described by the following model,

$$y^{(n)}(t) + \sum_{i=1}^n \left[a_i y^{(n-i)}(t) + b_i f_i(y^{(n-i)}) \right] = cu(t) \quad (1)$$

The system order **should be at least 3** (i.e., $n \geq 3$), the functions $f_i(x, t)$ are nonlinear, and the coefficients a_i, b_i are different from zero. The model can be taken from a scientific article, a textbook, or other scientific sources (please add a citation). You are required to explain the model's development for the selected process.

2) For the model (1), an adaptive controller has been presented in class (see slides on Adaptive-Control) to track a reference signal $y_d(t)$. Assume here that only a_i, b_i are unknown and suggest an alternative adaptive controller for (1) (for any n), designed to track the output $y_m(t)$ of a reference model. Use the classical MRAC approach, as was presented in class for first-order systems, where the controller gains are determined directly by the adaptation laws. Note that the reference model is an integral element of the complete control system. The control goal is $y(t) \rightarrow y_m(t)$ (for a given $r(t)$, which is the input of the reference model). Present the development of the adaptive controller and support your design with a stability proof.

3) According to the presented solution of 2), show the nominal control signal to be applied under the assumption that all model parameters are known. Use MATLAB/Simulink or Python to perform a simulation of the closed-loop system and present the simulation results with relevant explanations (which include explanations of the way the model is implemented by the software).

4) Assume that the model parameters a_i, b_i are unknown. Realize the adaptation law with the simulation of 3) and perform simulations to examine the performance of the adaptation law. Compare these results to those obtained in 3). Present a discussion to explain the implementation method and results.

5) Implementation of the control law (developed in 2) by a microcontroller requires digital implementation (based on a discrete-time formulation). Using the indirect digital control design approach, present a discrete-time model (based on difference equations) of the controllers developed in 2). Explain your design and implement it in MATLAB/Simulink or Python to simulate the closed loop (combining the discrete-time controller and the continuous-time process). Present a discussion on the implementation method.

6) Using simulations of the discrete-time controller, demonstrate the expected performance degradation associated with an increase in sampling time. Explain this phenomenon and suggest a way to reduce the performance degradation caused by the discrete-time realization. Present a new design with the proposed solution and present comparative simulations along with a discussion of the results.

General instructions

The work can be done in pairs or alone. The solution should be submitted as a printed document, including full documentation of all required computer work (MATLAB/Simulink or Python). The document language can be English or Hebrew. Please submit it to the Moodle submission box by June 28. Zoom meetings to defend your work will be scheduled after submission.